

BULLET-LA FRETAZ, SwitzerlandWMO: **066190**Lat: **46.8406N**Lon: **6.5764E**Elev: **3970**StdP: **12.71**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.2	14.3	1.1	6.7	23.8	6.3	8.8	23.7	25.0	35.2	20.8	32.9	6.2	60	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	75.3	62.2	72.5	61.1	69.8	59.9	64.1	72.2	62.5	70.2	61.0	68.3	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.1	93.1	68.5	59.4	87.6	66.8	57.8	82.9	65.2	31.5	72.6	30.2	70.4	29.0	68.4	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	14.1	11.8	8.5	80.0	5.0	2.9	4.9	82.2	1.9	83.9	-0.9	85.5	-4.5	87.7
			7.9	67.7	4.9	1.9	4.4	69.1	1.5	70.2	-1.2	71.3	-4.8	72.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	29.8	29.6	35.6	42.6	48.2	56.1	59.4	58.4	52.1	46.3	37.4	32.2	(6)
(7)		DBStd	12.95	7.65	8.70	8.03	7.62	7.20	7.32	6.45	6.63	6.21	7.55	8.11	7.94	(7)
(8)		HDD50	3184	626	572	448	241	122	24	6	7	48	157	380	553	(8)
(9)		HDD65	7696	1091	992	913	671	520	278	195	225	388	579	827	1017	(9)
(10)		CDD50	1015	0	0	1	20	67	207	297	267	111	43	2	0	(10)
(11)		CDD65	54	0	0	0	0	1	12	21	20	0	0	0	0	(11)
(12)		CDH74	148	0	0	0	0	1	34	58	55	0	0	0	0	(12)
(13)		CDH80	9	0	0	0	0	0	2	4	3	0	0	0	0	(13)
(14)	Wind	WSAvg	4.7	4.9	5.2	5.3	5.1	4.9	4.6	4.5	4.1	4.3	4.2	4.8	5.0	(14)
(15)	Precipitation	PrecAvg	50.6	4.0	3.7	3.5	3.8	4.4	4.3	4.5	4.3	4.2	4.5	4.6	4.8	(15)
(16)		PrecMax	64.3	10.8	10.1	13.3	6.7	7.4	9.4	9.7	8.7	9.4	8.3	12.7	10.0	(16)
(17)		PrecMin	36.1	0.7	0.7	0.9	0.6	1.5	1.9	2.0	1.1	1.6	0.9	0.5	0.0	(17)
(18)		PrecStd	7.5	2.4	2.2	2.7	1.9	1.8	1.9	1.9	1.9	2.0	2.0	2.8	2.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	51.2	55.4	64.1	70.1	77.9	79.5	79.3	69.8	64.6	56.7	50.9	(19)
(20)			MCWB	37.5	39.6	44.7	51.1	58.4	64.3	63.2	62.5	59.6	53.9	47.0	38.6	(20)
(21)		2%	DB	45.8	46.8	52.0	60.0	65.8	74.0	75.2	74.8	66.5	61.2	53.6	47.2	(21)
(22)			MCWB	37.9	37.1	42.5	48.7	55.9	62.4	61.7	62.0	58.5	52.0	45.1	38.1	(22)
(23)		5%	DB	42.6	43.4	49.2	56.8	62.6	70.6	72.1	71.8	63.6	58.6	50.8	44.7	(23)
(24)			MCWB	37.6	36.1	41.4	47.4	53.3	60.5	61.0	61.0	56.9	51.0	43.8	38.2	(24)
(25)		10%	DB	40.1	40.5	46.5	53.8	59.3	67.4	69.2	68.6	61.1	56.0	48.4	42.3	(25)
(26)			MCWB	36.2	35.7	39.8	45.9	51.6	58.8	59.6	59.8	55.4	49.9	42.8	37.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	41.9	47.0	52.4	60.1	67.1	65.6	65.9	62.2	56.4	50.0	44.1	(27)
(28)			MCDWB	44.8	46.3	51.3	60.6	67.7	75.2	74.3	74.1	66.9	61.6	53.9	45.8	(28)
(29)		2%	WB	40.8	39.6	44.5	50.1	57.1	63.8	63.6	63.8	59.7	54.2	47.0	41.8	(29)
(30)			MCDWB	42.8	43.6	50.0	58.3	64.3	71.7	72.0	71.8	64.7	58.1	50.8	44.1	(30)
(31)		5%	WB	38.9	38.0	42.7	48.1	54.5	61.5	62.0	62.1	57.8	52.8	45.3	40.2	(31)
(32)			MCDWB	41.2	41.8	47.7	55.3	60.8	68.9	70.2	69.6	62.5	56.4	49.1	42.7	(32)
(33)		10%	WB	36.7	36.2	40.8	46.2	52.3	59.4	60.5	60.5	56.1	51.2	43.7	38.5	(33)
(34)			MCDWB	39.5	39.9	45.3	53.0	58.3	66.3	68.1	67.4	60.5	54.9	47.3	41.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	8.7	9.9	11.6	11.9	12.6	12.9	11.9	10.7	9.8	8.6	8.4	(35)
(36)			MCDBR	10.7	11.6	13.1	14.8	15.6	15.2	15.5	14.7	13.3	12.8	11.4	10.7	(36)
(37)			MCWBR	7.3	8.0	9.1	9.3	10.5	10.4	9.2	9.5	9.4	9.2	8.0	7.2	(37)
(38)		5% WB	MCDWB	9.4	10.7	11.6	13.9	14.4	14.5	14.4	13.9	12.0	10.9	10.4	9.3	(38)
(39)			MCWBR	7.7	8.4	8.6	9.3	10.1	10.4	9.2	9.4	9.0	8.7	8.3	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.262	0.282	0.312	0.342	0.350	0.352	0.351	0.345	0.328	0.312	0.284	0.261	(40)	
(41)		taud	2.488	2.450	2.439	2.381	2.393	2.415	2.405	2.437	2.472	2.512	2.535	2.502	(41)	
(42)		Ebn at Noon	271	285	291	290	290	289	287	285	282	270	259	258	(42)	
(43)		Edh at Noon	20	26	30	36	36	36	36	33	29	24	19	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	405	676	1044	1383	1588	1881	1834	1597	1240	795	438	329	(44)	
(45)		RadStd	48	100	153	200	144	176	160	147	111	79	57	56	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (94 neighbors)	N/A	N/A	N/A	+0.70	+0.52	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page